Consider, (–7) × (–42),

$$(-7) \times (-42) = +294$$

Therefore, (–42) × (–7) and (–7) × (–42) are equal.

It is commutative.

Example 1.22

Prove that [(–2) × 3] ×(–4) = (–2) ×[3×(–4)].

Solution

In the first case (–2) and (3) are grouped together and in the second case (3) and (–4) are grouped together

L.H.S = [(–2) × 3] ×(–4)

= (–6) × (–4) = 24

R.H.S = (–2) ×[3×(–4)]

= (–2) × (–12) = 24

Therefore, L.H.S. = R.H.S.

[(–2) × 3] ×(–4) = (–2) ×[3×(–4)] Hence it is proved.

Example 1.23

Are (–81) × [5×(–2)] and [(–81) × 5]×(–2) equal? Mention the property.

Solution

Consider, (–81) × [5×(–2)],

$$(-81) \times [5\times(-2)] = (-81) \times (-10) = 810$$

Consider, [(–81) × 5]×(–2),

$$[(-81) \times 5]\times(-2) = (-405) \times (-2) = 810$$

Therefore, (–81) × [5×(–2)] and [(–81) × 5]×(–2) are equal.

It is associative.

Example 1.24

Are 3 × [(–4)+6] and [3 ×(–4)]+(3×6) equal? Mention the property.

Solution

Consider, 3 × [(–4)+6],

$$3 \times [(-4)+6] = 3 \times 2 = 6$$

Consider, [3 ×(–4)]+[3×6],

$$[3 \times(-4)]+[3\times6] = -12+18 = 6$$

Therefore, 3 × [(–4)+6] and [3 ×(–4)]+3×6 are equal.

It is the distributive property of multiplication over addition.

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